P1-04 — Formal Proof Package for T1–T4 (single component)

Command: P1-04 — Formal Proof Package for T1–T4 **Date:** 21 October 2026 **Scope:** Single component only. Complete proofs of RR-1, RR-2, RR-3, RR-4, RR-5 and of Theorems T1–T4 under the convention lock A1–A10 of P1_03_MATHEMATICAL_PROOF_PLAN.md. **Frozen statements:** 02_THEOREM_BLUEPRINT.md is **not** edited. PC-1…PC-6 remain **unapproved** proposals (register: 04_BLUEPRINT_GAP_REGISTER.md, GAP-17).

Label note (no silent renumbering). This command calls the hull generator/dimension identity "RR-4". In the frozen blueprint, RR-4 is the twist-transport check and RR-5 is the lcm lemma. Both are proved below; the hull assembly is labelled **RR-4 (command) = blueprint RR-4 + RR-5 → T4**. Blueprint labels are otherwise kept.

Status of this file: the proofs below are complete and self-contained under the stated hypotheses. They are a proof package, not manuscript prose. Computational cross-checks are collected in §9 and are supporting evidence only.

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0. Standing hypotheses and the dual-order convention (verbatim)

Throughout, one component is fixed and the subscript s is suppressed.

- **(S1)** p is prime, K = F_{p^d}, and k is an integer with $0 \le k \le d-1$.

$\sigma(\alpha) = \alpha^{p^k}$ and $\rho(\alpha) = \alpha^{p^{d-k}}$ for $\alpha \in K$. Since $\alpha^{p^d} = \alpha$ on K, $\rho = \sigma^{-1}$ on K.

- **(S2)** $\varrho \ge 0$, $P = p^\varrho \ge 1$, $n = P \cdot n_0$ with $\gcd(n_0, p) = 1$.
- **(S3)** $\lambda \in K^*$ satisfies the compatibility condition $\lambda^{1 + p^{d-k}} = 1$, and $\mu \in K^*$ is the unique element with $\mu^P = \lambda$ (the map $a \mapsto a^P$ is a bijection of the finite field K).
- **(S4)** $M(x) = x^n - \lambda$, $S(x) = x^{n_0} - \mu$. Then $M = S^P$ (characteristic p: $(x^{n_0} - \mu)^P = x^{n_0 P} - \mu^P$), and S is separable with distinct monic irreducible factors: $S = \prod_{f \in F} f$, $M = \prod_{f \in F} f^P$. (Separability: $S' = n_0 x^{n_0 - 1}$, $p \nmid n_0$, and every root α of S has $\alpha^{n_0} = \mu \ne 0$, hence $\alpha \ne 0$ and $S'(\alpha) \ne 0$.)
- **(S5)** Codes are K-linear subspaces of K^n, identified with polynomials of degree $< n$; a λ-constacyclic code is an ideal of $R = K[x]/(M)$. For $a = (a_f)_{f \in F} \in \{0,\dots,P\}^F$, $g_a = \prod_{f \in F} f^{a_f}$ and $C(a) = \langle g_a \rangle \subseteq R$; $h_a = M / g_a = \prod_{f \in F} f^{P - a_f}$.
- **(S6)** For monic $f(x) = \Sigma_{i=0}^r c_i x^i \in K[x]$ with $c_0 \ne 0$ (automatic for $f \in F$, since $f \mid S$ and $S(0) = -\mu \ne 0$), the reciprocal is

$f^\#(x) = \rho(c_0)^{-1} \Sigma_{i=0}^r \rho(c_i) x^{r-i}$, and $\tau(f) = f^\#$.

- **(S7) Dual-order convention (stated verbatim, as required):**

"The code-first pairing is $\langle x, y \rangle_k = \Sigma_{i=0}^{n-1} x_i \sigma(y_i)$, and the code-first dual is $D(C) = \{ y \in K^n : \langle c, y \rangle_k = 0 \text{ for all } c \in C \}$ — the code vector occupies the first slot and the candidate vector the second slot. The Euclidean dual is $C^\perp = \{ z \in K^n : \Sigma_i c_i z_i = 0 \text{ for all } c \in C \}$, and $\mathrm{Hull}_k(C) = C \cap D(C)$." With this order, §5 shows $D(C) = \rho(C^\perp)$ (coordinatewise ρ). Under the opposite order the roles of σ and ρ would interchange; that reading is **not** used anywhere in this file.

- **(S8)** A τ-orbit is enumerated $O = (f_0, \dots, f_{L-1})$ with $\tau(f_i) = f_{i+1}$ (indices mod L), and $\alpha_i := a_{f_i}$. The predecessor of i is $i-1$.

Dimensions are over K unless stated otherwise. If the component is read as the blueprint's $K_s = F_{q^{m_s}}$, then $\dim_{F_q} = m_s \cdot \dim_K$ for every K-subspace below, because $[K : F_q] = m_s$.

Two foundational facts about K[x] are used and proved here, so that nothing is hidden behind the word "standard":

Fact 0.1 (division; K[x] is a principal ideal domain). For $\varphi, \psi \in K[x]$, $\psi \ne 0$, there exist unique q, r with $\varphi = q\psi + r$ and $r = 0$ or $\deg r < \deg \psi$ (polynomial long division; the leading coefficient of ψ is invertible because K is a field). Consequently every ideal $I \ne (0)$ of K[x] is principal, generated by any

monic element of I of minimal degree: if φ ∈ I, divide φ = qḡ + r; then r = φ − qḡ ∈ I and minimality forces r = 0.

Fact 0.2 (unique factorization). Every nonconstant φ ∈ K[x] factors into monic irreducibles (induction on deg φ: either φ is irreducible or φ = φ_1 φ_2 with both degrees smaller), and the factorization is unique up to order. Uniqueness uses Euclid's lemma: if π is irreducible and π | uv, then π | u or π | v. Indeed, by Fact 0.1 the ideal (π, u) is generated by a common divisor of π and u; since π is irreducible this generator is 1 or an associate of π. If π ∤ u, the generator is 1, so Aπ + Bu = 1 for some A, B; multiplying by v gives v = Aπv + B(uv), and π divides both terms on the right, hence π | v. Uniqueness then follows by cancelling one irreducible factor at a time (induction on the number of factors). For π irreducible, the valuation v_π(φ) is the exponent of π in this factorization; φ | ψ iff v_π(φ) ≤ v_π(ψ) for all π.

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1. RR-1 — Divisor/exponent correspondence and the dimension formula

Lemma RR-1. Let M ∈ K[x] be monic of degree n ≥ 1 with M = ∏_{f∈F} f^{P} the factorization of Fact 0.2 into distinct monic irreducibles (P ≥ 0 an integer; in our setting P = p^ϱ ≥ 1).

(i) The map a ↦ g_a = ∏_{f∈F} f^{a_f} is a bijection from {0,...,P}^F onto the set of monic divisors of M.

(ii) The map a ↦ ⟨g_a⟩ ⊆ K[x]/(M) is a bijection from {0,...,P}^F onto the set of ideals of K[x]/(M).

(iii) For every monic divisor g of M, dim_K ⟨g⟩ = n − deg g (the ideal read as the set of multiples of g of degree < n).

Proof. (i) If a ∈ {0,...,P}^F, then g_a | M with v_f(g_a) = a_f ≤ P = v_f(M) for f ∈ F and v_π(g_a) = 0 = v_π(M) for π ∉ F, by Fact 0.2. Conversely, if g | M is monic, then v_f(g) ≤ v_f(M) = P for f ∈ F and v_π(g) = 0 for π ∉ F, so g = ∏ f^{v_f(g)} = g_a for a_f := v_f(g). The two constructions are mutually inverse because Fact 0.2 makes the exponents v_f(g) intrinsic to g.

(ii) Ideals of K[x]/(M) correspond bijectively to ideals I of K[x] with (M) ⊆ I (preimage under the quotient map; this is the definition of the quotient ring's ideals). By Fact 0.1, I = (ḡ) for a unique monic ḡ (the monic generator is unique: two generators differ by a unit of K[x], i.e. a nonzero constant, and only one is monic). The inclusion (M) ⊆ (ḡ) holds iff ḡ | M. Hence ideals of the quotient correspond to monic divisors of M, and (i) finishes the proof. No hypothesis on M (in particular no separability or square-freeness) is used.

(iii) Let g | M be monic, r = deg g, and put h = M/g (a polynomial, of degree n − r). Consider the K-linear map Φ : K[x]/(h) → (g)/(M), [u] ↦ [gu]. It is well defined (if u − u′ ∈ (h) then g(u − u′) ∈ (gh) = (M)), and it is surjective by the definition of (g)/(M). It is injective: if gu ∈ (M) = (gh), then gh | gu in K[x], and cancelling g (Fact 0.2, or directly: gu = gh·w implies u = hw since K[x] is a domain) gives u ∈ (h). Hence Φ is an isomorphism. Now dim_K K[x]/(h) = deg h = n − r: the classes of 1, x, ..., x^{deg h − 1} form a basis, because every polynomial is congruent modulo h to its remainder of degree < deg h (Fact 0.1, existence), and a polynomial of degree < deg h divisible by h is zero (Fact 0.1, degree bound). Therefore dim_K ⟨g⟩ = n − deg g. ∎

Theorem T1 (parametrization, single component). Under (S1)–(S5), the map a ↦ C(a) = ⟨g_a⟩ is a bijection from {0,...,P}^F onto the set of λ-constacyclic codes of length n over K, and dim_K C(a) = n − Σ_{f∈F} a_f·deg f. In particular there are (P+1)^{|F|} such codes.

Proof. By (S5), λ-constacyclic codes are exactly the ideals of R = K[x]/(M). Lemma RR-1(ii) with M = ∏ f^P (S4) gives the bijection, and RR-1(iii) gives dim_K C(a) = n − deg g_a = n − Σ_f a_f deg f. ∎

Hypotheses actually used: Fact 0.1–0.2 and M = ∏ f^P. Neither gcd(n_0, p) = 1 nor compatibility (S3) is used in T1; they enter through (S4)'s guarantee that the distinct factors of M are exactly the factors of the separable polynomial S, each with multiplicity exactly P.

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2. RR-2 — Coefficientwise ρ and the reciprocal

Write $\bar\rho : K[x] \to K[x]$, $\bar\rho(\Sigma_i c_i x^i) = \Sigma_i \rho(c_i) x^i$, and let $\bar\sigma$ be the analogous map for σ.

Lemma RR-2. (i) $\bar\rho$ is a ring automorphism of $K[x]$ with inverse $\bar\sigma$. (ii) For monic f of degree r with $f(0) \neq 0$, f# (S6) is monic of degree r with $f^\#(0) \neq 0$; explicitly, if $f^\natural_\rho := \rho(c_0)^{-1} \cdot x^r \bar\rho(f)(1/x)$, then $f^\natural_\rho = f^\#$. (iii) # is multiplicative on monic polynomials with nonzero constant term: $(fg)^\# = f^\# \cdot g^\#$. Hence $(\prod_{f\in F} f^{a_f})^\# = \prod_{f\in F} (f^\#)^{a_f}$ for any exponents $a_f \geq 0$. (iv) (Root transport.) Let f be monic irreducible of degree r with $f(0) \neq 0$, and let $\alpha \in \bar K$ (an algebraic closure of K) with $f(\alpha) = 0$. Then f# is the minimal polynomial over K of $\beta := \alpha^{-p^{d-k}}$; in particular f# is monic irreducible of degree r and $f^\#(\beta) = 0$. (v) (Compatibility, two forms.) $\lambda^{1 + p^{d-k}} = 1 \iff \mu^{1 + p^{d-k}} = 1$. (vi) Under (S3)–(S4), $\tau : f \mapsto f^\#$ maps F into F and is a permutation of F; all factors in one τ-orbit have the same degree.

Proof. (i) $\bar\rho$ preserves addition coefficientwise and preserves multiplication because ρ is a field homomorphism: the coefficient of x^m in $\bar\rho(\varphi\psi)$ is $\rho(\Sigma_{i+j=m} \varphi_i \psi_j) = \Sigma_{i+j=m} \rho(\varphi_i)\rho(\psi_j)$, which is the coefficient of x^m in $\bar\rho(\varphi)\bar\rho(\psi)$. Since $\rho\sigma = \sigma\rho = \mathrm{id}_K$ (S1), $\bar\sigma\bar\rho = \bar\rho\bar\sigma = \mathrm{id}$, so $\bar\rho$ is an automorphism with inverse $\bar\sigma$.

(ii) With $f = \Sigma_{i=0}^r c_i x^i$, $c_r = 1$: $x^r \bar\rho(f)(1/x) = \Sigma_{i=0}^r \rho(c_i) x^{r-i}$. Its coefficient of x^r is $\rho(c_0)$; multiplying by $\rho(c_0)^{-1}$ gives exactly (S6). The result is monic (leading coefficient $\rho(c_0)^{-1}\rho(c_0) = 1$), of degree r (leading coefficient nonzero), and its constant term is $\rho(c_0)^{-1}\rho(c_r) = \rho(c_0)^{-1} \neq 0$.

(iii) Let f, g be monic of degrees r, t with nonzero constant terms. Applying the ring automorphism $\bar\rho$ to the product gives $\bar\rho(fg) = \bar\rho(f)\bar\rho(g)$. Reversal is multiplicative on polynomials of fixed degrees: $x^{r+t}(\bar\rho(fg))(1/x) = [x^r \bar\rho(f)(1/x)]\cdot[x^t \bar\rho(g)(1/x)]$, because both sides equal $x^{r+t}\bar\rho(f)(1/x)\bar\rho(g)(1/x)$. The normalizing constants also multiply: $(fg)_0 = f_0 g_0$, so $\rho((fg)_0)^{-1} = \rho(f_0)^{-1}\rho(g_0)^{-1}$. Combining the three identities and using (ii) gives $(fg)^\# = f^\# g^\#$. The displayed consequence follows by induction on the number of factors (with $a_f = 0$ contributing the empty factor 1, and $(f^a)^\# = (f^\#)^a$ by induction on a).

(iv) Let $\Phi : \bar K \to \bar K$, $\Phi(t) = t^{p^{d-k}}$. Then Φ is a field endomorphism of $\bar K$ (Frobenius in characteristic p), injective (a field homomorphism out of a field is injective or zero, and $\Phi(1) = 1$), and surjective because $\bar K$ is algebraically closed, so $X^{p^{d-k}} - t$ has a root in $\bar K$ for every t; hence Φ is an automorphism of $\bar K$. On $K = F_{p^d}$, $\Phi(c) = c^{p^{d-k}} = \rho(c)$. Note $\alpha \neq 0$ because $f(0) \neq 0$, so $\beta = \Phi(\alpha)^{-1}$ is defined. First, $f^\#(\beta) = 0$: applying Φ to $f(\alpha) = \Sigma\, c_i \alpha^i = 0$ gives $\Sigma\, \rho(c_i)\, \Phi(\alpha)^i = 0$, i.e. $\Sigma\, \rho(c_i)\, \beta^{-i} = 0$; multiplying by β^r gives $\Sigma\, \rho(c_i)\, \beta^{r-i} = 0$, which is $\rho(c_0)\cdot f^\#(\beta) = 0$ by (ii), and $\rho(c_0) \neq 0$. Next, degrees: Φ maps $K(\alpha)$ onto $K(\Phi(\alpha))$ (it maps the coefficient field K onto itself and the generator α to $\Phi(\alpha)$), and it preserves the degree: if $\Sigma_{i=0}^{r-1} e_i \Phi(\alpha)^i = 0$ with $e_i \in K$, write $e_i = \Phi(d_i)$ with $d_i = \sigma(e_i) \in K$ and apply Φ^{-1} to get $\Sigma\, d_i \alpha^i = 0$, so every $d_i = 0$ because $[K(\alpha) : K] = r$ (f is the minimal polynomial of α, being monic irreducible with $f(\alpha) = 0$); spanning is identical. Hence $[K(\Phi(\alpha)) : K] = r$, and $K(\beta) = K(\Phi(\alpha))$ (as $\beta = \Phi(\alpha)^{-1}$), so the minimal polynomial m_β of β over K has degree r. Since $m_\beta \mid f^\#$ (because $f^\#(\beta) = 0$) and both are monic of degree r, $m_\beta = f^\#$. In particular f# is irreducible.

(v) Put $D = p^{d-k}$. Since $\lambda = \mu^P$ and the exponents commute ($P\cdot(1 + D) = (1 + D)\cdot P$ as integers), $(\mu^{1+D})^P = \mu^{P(1+D)} = (\mu^P)^{1+D} = \lambda^{1+D}$. Therefore $\lambda^{1+D} = 1 \iff (\mu^{1+D})^P = 1$. In a field of characteristic p, $y^P = 1$ implies $y = 1$, because $y^P - 1 = (y - 1)^P$ (the binomial coefficients $C(P, i)$, $0 < i < P$, are divisible by p as P is a power of p) and a field has no nilpotents. Hence $(\mu^{1+D})^P = 1 \iff \mu^{1+D} = 1$, proving the equivalence.

(vi) Let $f \in F$ and α a root of f in $\bar K$; then $\alpha^{n_0} = \mu$ (as $f \mid S$). By (iv), $\tau(f) = f^\#$ is the minimal polynomial of $\beta = \Phi(\alpha)^{-1}$ over K. Moreover $\beta^{n_0} = \Phi(\alpha)^{-n_0} = \Phi(\alpha^{n_0})^{-1} = \rho(\mu)^{-1}$. By (S3) and (v), $\mu^{1 + p^{d-k}} = 1$, i.e. $\rho(\mu) = \mu^{p^{d-k}} = \mu^{-1}$, so $\beta^{n_0} = \mu$. Thus β is a root of the separable polynomial S, so its minimal polynomial $\tau(f)$ divides S (a minimal polynomial of a root of S divides S, and separability is what makes the factors of S distinct); since $\tau(f)$ is monic irreducible (iv), $\tau(f) \in F$. So $\tau : F \to F$. For bijectivity, define τ' exactly as τ but with σ in place of ρ (i.e. (S6) with $\sigma(c_0)^{-1}\ \Sigma\ \sigma(c_i)\ x^{r-i}$). The argument of (iv) with $\Psi(t) =$

t^{p^k} (extending σ) shows τ′(g) is the minimal polynomial of γ^{−p^k} for any root γ of g. Taking g = τ(f) and γ = β, we get that τ′τ(f) is the minimal polynomial of β^{−p^k} = Φ(α)^{p^k} = α^{p^{d−k}·p^k} = α^{p^d}. Since K = F_{p^d}, the map t ↦ t^{p^d} fixes K pointwise, maps α to another root of f, and preserves degrees exactly as in (iv); hence the minimal polynomial of α^{p^d} over K is f itself. Therefore τ′τ(f) = f for all f ∈ F; the symmetric computation gives ττ′(g) = g. So τ is a permutation of the finite set F with inverse τ′. Finally, deg τ(f) = deg f by (ii), so all factors in a τ-orbit have equal degree. ∎

Theorem T2 (reciprocal, single component). Under (S1)–(S6): (i) # is multiplicative on monic polynomials with nonzero constant term, and in particular (∏_{f∈F} f^{a_f})# = ∏_{f∈F} (f#)^{a_f}; (ii) if f(α) = 0 for α ∈ \bar K, α ≠ 0, then f#(α^{−p^{d−k}}) = 0; (iii) λ^{1+p^{d−k}} = 1 ⟺ μ^{1+p^{d−k}} = 1, and under (S3) the map τ = # permutes F preserving degrees.

Proof. (i) is Lemma RR-2(iii); (ii) is RR-2(iv) (with α ≠ 0 guaranteed when f(0) ≠ 0); (iii) is RR-2(v)–(vi). ∎

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3. RR-3 — The Euclidean-dual generator lemma (no separability)

This lemma uses none of the repeated-root structure. We isolate exactly what it needs.

Lemma RR-3. Let n ≥ 1, λ ∈ K∗, and let g ∈ K[x] be a monic divisor of x^n − λ (no condition on n, on g, or on separability). Put r = deg g, h = (x^n − λ)/g, s = deg h = n − r, and let C = ⟨g⟩ = {gu : deg(gu) < n} ⊆ K^n under the coefficient identification. Define the normalized reciprocal h^♮(x) = h_0^{-1} Σ_{i=0}^s h_i x^{s−i} where h = Σ h_i x^i (note h_0 ≠ 0, proved below). Then the Euclidean dual satisfies C^⊥ = ⟨h^♮⟩ = {h^♮ v : deg(h^♮ v) < n}, and h^♮ is a monic divisor of x^n − λ^{-1}; i.e. C^⊥ is the λ^{-1}-constacyclic code with generator h^♮.

Proof. Since gh = x^n − λ, comparing constant terms gives g_0 h_0 = −λ ≠ 0; hence h_0 ≠ 0 (and g_0 ≠ 0), so h^♮ is well defined, monic of degree s.

Dimension preliminaries. The shifts g, xg, ..., x^{n−r−1}g have distinct degrees r, r+1, ..., n−1, hence are linearly independent, and every multiple of g of degree < n is a K-combination of them; so dim_K C = n − r. Identically, the multiples of h^♮ of degree < n form a space of dimension n − s = r. Also, dim_K C^⊥ = n − dim_K C = r: indeed, if B is a basis of C written as rows of a (dim C) × n matrix, then C^⊥ is the kernel of the linear map K^n → K^{dim C}, z ↦ (Σ_i c_i z_i)_{c ∈ B}; that map is surjective (its rows are linearly independent) and its kernel is exactly C^⊥ (orthogonality to a basis is orthogonality to C), so rank–nullity gives dim C^⊥ = n − dim C.

∗The candidate divides x^n − λ^{-1}.∗ For a polynomial φ of degree t, write φ^∗(x) = x^t φ(1/x) (unnormalized reciprocal). If deg φ_1 = t_1, deg φ_2 = t_2, then (φ_1 φ_2)^∗ = φ_1^∗ φ_2^∗ (both sides equal x^{t_1 + t_2} φ_1(1/x) φ_2(1/x)). Applying this to gh = x^n − λ: g^∗ h^∗ = (x^n − λ)^∗ = x^n((1/x)^n − λ) = 1 − λx^n = −λ(x^n − λ^{-1}). Hence h^∗ | x^n − λ^{-1} (the factor −λ ∈ K∗ is a unit), and h^♮ = h_0^{-1} h^∗ is a unit multiple of h^∗, so h^♮ | x^n − λ^{-1} as well.

∗Orthogonality (inclusion ⟨h^♮⟩ ⊆ C^⊥).∗ Two trivial cases first. If r = 0, then g = 1, C = K^n, C^⊥ = {0}, and h^♮ has degree s = n, whose only multiple of degree < n is 0; equality holds. If s = 0, then h = 1, g = x^n − λ, C = {0}, C^⊥ = K^n, and h^♮ = 1, whose multiples of degree < n are all of K^n; equality holds. Assume now r, s ≥ 1.

For a, d ∈ K[x] of degree < n, define the reversal rev_n(d)(x) = x^{n−1} d(1/x) = Σ_{i=0}^{n−1} d_i x^{n−1−i}. Then the Euclidean pairing is a single coefficient:

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⟨a, d⟩_E := Σ_{i=0}^{n−1} a_i d_i = [x^{n−1}] ( a(x)·rev_n(d)(x) ),        (1)
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because the coefficient of x^{n−1} in a(x)·rev_n(d)(x) is Σ_i a_i · [x^{n−1−i}] rev_n(d) = Σ_i a_i d_i.

Take a ∈ C and d ∈ ⟨h^♮⟩; write a = gu with deg u < s (possible since deg a < n = r + s and g | a) and d = h^♮ v with deg v < r. Put V(x) = x^{r−1} v(1/x) (a polynomial of degree ≤ r − 1, since deg v ≤ r − 1; if v = 0 then d = 0 and there is nothing to prove). We compute rev_n(d). First, h^♮(1/x) = h_0^{-1} Σ_i h_i (1/x)^{s−i} = h_0^{-1} x^{−s} Σ_i h_i x^i = h_0^{-1} x^{−s} h(x). Therefore

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rev_n(d)(x) = x^{n−1} h^↯(1/x) v(1/x)
            = h_0^{-1} x^{n−1−s} h(x) v(1/x)
            = h_0^{-1} x^{r−1} h(x) · x^{−(r−1)} V(x)        (since n − 1 − s = r − 1)
            = h_0^{-1} h(x) V(x).                            (2)
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Hence, using a = gu and gh = x^n − λ,

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a(x)·rev_n(d)(x) = h_0^{-1} (x^n − λ)·(u(x) V(x)).           (3)
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Now deg u ≤ s − 1 and deg V ≤ r − 1, so w := uV satisfies deg w ≤ n − 2 (or w = 0). In (x^n − λ)w, the term x^n w has only degrees ≥ n (or vanishes), and the term −λw has only degrees ≤ n − 2; therefore the coefficient of x^{n−1} is 0. By (1), ⟨a, d⟩_E = 0. Since a ∈ C and d ∈ ⟨h^↯⟩ were arbitrary, ⟨h^↯⟩ ⊆ C^⊥.

Equality. Both spaces have dimension r (dimension preliminaries), and one is contained in the other; finite-dimensional spaces with an inclusion and equal dimension are equal. Hence C^⊥ = ⟨h^↯⟩. ∎

Conditions used by RR-3 (complete list): λ ∈ K∗; g monic with g | x^n − λ; the coefficient identification of codes with polynomials of degree < n. Nothing else: in particular **not** gcd(n, p) = 1, **not** n = Pn_0, **not** compatibility, **not** separability of x^n − λ. The proof uses only the identity gh = x^n − λ, the reversal identity (1)–(2), the degree bounds deg u < s, deg v < r, and dimension counts.

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4. Transport by ρ, and RR-4 (command): the Galois dual generator

Lemma 4.1 (transport of codes by a coefficient automorphism). Let φ ∈ {σ, ρ} and let φ̄ be the coefficientwise map of Lemma RR-2(i) (for φ = ρ) resp. its analogue for σ. If E = ⟨e⟩ ⊆ K[x]/(x^n − η) with e a monic divisor of x^n − η, then φ̄(E) = ⟨φ̄(e)⟩ ⊆ K[x]/(x^n − φ(η)).

Proof. φ̄ is a ring automorphism of K[x] (Lemma RR-2(i) for ρ; the same coefficient computation for σ), and φ̄(x^n − η) = x^n − φ(η) because φ̄ fixes 1 and the indeterminate's coefficients. Hence φ̄ maps the ideal (e) onto (φ̄(e)) and the ideal (x^n − η) onto (x^n − φ(η)), inducing the claimed identification of quotient ideals. Elementwise: d = ev with deg d < n implies φ̄(d) = φ̄(e)φ̄(v) with deg φ̄(v) = deg v (a field automorphism sends the nonzero leading coefficient of v to a nonzero element), so multiples of degree < n correspond exactly. ∎

Proposition 4.2 (blueprint RR-4: twist transport). For any K-linear code C ⊆ K^n, with D(C) and C^⊥ as in (S7),

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D(C) = ρ(C^⊥)   (coordinatewise ρ applied to the Euclidean dual).
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Consequently, if C = ⟨g⟩ ⊆ K[x]/(x^n − λ) and h = (x^n − λ)/g, then D(C) = ⟨h#⟩ ⊆ K[x]/(x^n − λ^{−p^{d−k}}). If moreover λ is compatible (S3), then λ^{−p^{d−k}} = λ, so D(C) = ⟨h#⟩ ⊆ K[x]/(x^n − λ).

Proof. By (S7), y ∈ D(C) iff for every c ∈ C, Σ_{i=0}^{n−1} c_i σ(y_i) = 0. Put w = σ(y) (coordinatewise σ); this says exactly Σ_i c_i w_i = 0 for all c ∈ C, i.e. w ∈ C^⊥. Since σ is a bijection of K^n with inverse ρ, y ∈ D(C) ⟺ σ(y) ∈ C^⊥ ⟺ y ∈ ρ(C^⊥). This proves the displayed identity — this is the exact point where the argument order in (S7) is used: the candidate sits in the σ-slot, so solving for the candidate applies σ^{-1} = ρ to the Euclidean dual.

Now apply Lemma RR-3: C^⊥ = ⟨h^↯⟩ ⊆ K[x]/(x^n − λ^{-1}). By Lemma 4.1 with φ = ρ, D(C) = ρ(C^⊥) = ⟨ρ̄(h^↯)⟩ ⊆ K[x]/(x^n − ρ(λ^{-1})). We identify the generator: ρ(λ^{-1}) = ρ(λ)^{-1} = (λ^{p^{d−k}})^{-1} = λ^{−p^{d−k}}, and ρ̄(h^↯) = ρ(h_0)^{-1} Σ_{i=0}^s ρ(h_i) x^{s−i} = h# by comparing with (S6) applied to h (which is legitimate: h is monic and h_0 ≠ 0 by RR-3). Finally, compatibility (S3) says λ^{1 + p^{d−k}} = 1, i.e. λ^{p^{d−k}} = λ^{-1}, hence λ^{−p^{d−k}} = λ. ∎

Theorem T3 (code-first dual, single component). Under (S1)–(S6), for every assignment a, with h_a = M/g_a,

```
D(C(a)) = ⟨h_a#⟩ ⊆ K[x]/(x^n − λ^{−p^{d−k}}),
```

and under compatibility (S3) D(C(a)) = ⟨h_a#⟩ ⊆ K[x]/(x^n − λ). The exponents of h_a# are: for each τ-orbit O = (f_0, …, f_{L−1}) (S8) and α_i = a_{f_i}, the exponent of f_i in h_a# is

```
β_i = P − α_{i−1}        (indices mod L).
```

Proof. Apply Proposition 4.2 to C = C(a), g = g_a, h = h_a (RR-3 applies by (S5): g_a is a monic divisor of x^n − λ and h_a = (x^n − λ)/g_a). This gives the generator statement and the twist statement. For the exponents: by (S5), h_a = ∏_{f∈F} f^{P − a_f}. By Theorem T2(i) and Lemma RR-2(vi) (τ permutes F), h_a# = ∏_{f∈F} (τ(f))^{P − a_f} = ∏_{g∈F} g^{P − a_{τ^{-1}(g)}} (reindexing g = τ(f)). Fix an orbit O and g = f_i. Then τ^{-1}(f_i) = f_{i−1} by (S8), so the exponent of f_i in h_a# is P − a_{f_{i−1}} = P − α_{i−1} =: β_i. ∎

Remark (what compatibility is and is not needed for). The identity D(C(a)) = ⟨h_a#⟩ and the exponent formula β_i = P − α_{i−1} hold **without** compatibility; compatibility is used only in the last sentence of Proposition 4.2 to identify the ambient modulus x^n − λ^{−p^{d−k}} with x^n − λ, i.e. to regard D(C(a)) as a λ-constacyclic code in the same quotient. This distinction matters in §6.

---

5. RR-5 — The lcm lemma in a PID and in the quotient

Lemma RR-5. (i) For nonzero g, l ∈ K[x], (g) ∩ (l) = (lcm(g, l)), where lcm is the monic least common multiple. (ii) Let g, l be monic divisors of a monic M ∈ K[x], and let π : K[x] → K[x]/(M) be the quotient map. Then π((g)) ∩ π((l)) = π((lcm(g, l))) (and lcm(g, l) | M). (iii) If g = ∏_π π^{u_π} and l = ∏_π π^{v_π} are the factorizations of Fact 0.2, then lcm(g, l) = ∏_π π^{max(u_π, v_π)}.

Proof. (iii) By Fact 0.2, a polynomial m is a common multiple of g and l iff v_π(m) ≥ u_π and v_π(m) ≥ v_π for every irreducible π, iff v_π(m) ≥ max(u_π, v_π) for all π. The monic polynomial with v_π = max(u_π, v_π) is therefore a common multiple dividing every common multiple; it is the lcm, which proves (iii) and also the divisibility characterization used next.

(i) x ∈ (g) ∩ (l) iff g | x and l | x, i.e. iff x is a common multiple of g and l. By the previous paragraph, the common multiples are exactly the multiples of lcm(g, l). Hence (g) ∩ (l) = (lcm(g, l)).

(ii) Put m = lcm(g, l). Since g | M and l | M, M is a common multiple of g and l, so m | M by the characterization in (iii). In particular (M) ⊆ (g), (M) ⊆ (l), and (M) ⊆ (m). We show the two sides of the claimed equality have the same elements. If y ∈ π((m)), write y = π(Y) with Y ∈ (m); since (m) ⊆ (g) and (m) ⊆ (l) (as g | m and l | m), y ∈ π((g)) ∩ π((l)). Conversely, let y ∈ π((g)) ∩ π((l)) and choose any representative Y ∈ K[x] with π(Y) = y. Membership y ∈ π((g)) means Y = G + Z with G ∈ (g) and Z ∈ (M); but (M) ⊆ (g), so Y ∈ (g). Identically, Y ∈ (l). Hence Y ∈ (g) ∩ (l) = (m) by (i), so y = π(Y) ∈ π((m)). ∎

Applicability (precise). Lemma RR-5(ii) needs exactly: K[x] a PID/UFD (Facts 0.1–0.2) and g, l both dividing the **same** modulus M. In the application below, g = g_a and l = h_a# both divide M = x^n − λ: for g_a by construction (S5), and for h_a# by Theorem T3 under compatibility. Without compatibility, h_a# divides x^n − λ^{−p^{d−k}} instead, and (ii) cannot be applied in the single quotient K[x]/(M) — see §6.

---

6. RR-4 (command) / Theorem T4 — Hull generator and dimension

Theorem T4 (hull, single component). Under (S1)–(S6) including compatibility (S3), for every assignment a:

```
Hull_k(C(a)) = ⟨ lcm(g_a, h_a#) ⟩ ⊆ K[x]/(x^n − λ),
```

the exponent of f_i in the hull generator is max(α_i, P − α_{i−1}) (orbit notation of (S8)), and

```
dim_K C(a)      = Σ_{O} deg(f_O) · Σ_{i=0}^{L_O−1} (P − α_i),
dim_K Hull_k(C(a)) = Σ_{O} deg(f_O) · Σ_{i=0}^{L_O−1} min(P − α_i, α_{i−1}),
```

where the outer sums are over all τ-orbits O of F, f_O is any factor in O (all have the same degree, Lemma RR-2(vi)), and L_O = |O|. The blueprint's dim_{F_q} formulas are these with deg(f_O) replaced by w_O = m_s·deg_{K_s}(f_O) and summed over components s.

Proof. By definition (S7), Hull_k(C(a)) = C(a) ∩ D(C(a)) as subspaces of K^n, i.e. as ideals of R = K[x]/(M). By Theorem T1, C(a) = π((g_a)); by Theorem T3 (compatible case), D(C(a)) = π((h_a#)), with

both g_a | M and h_a# | M. Lemma RR-5(ii) therefore gives Hull_k(C(a)) = π((g_a)) ∩ π((h_a#)) = π((lcm(g_a, h_a#))) = ⟨lcm(g_a, h_a#)⟩. By Lemma RR-5(iii) and the exponent computations of (S5) and Theorem T3, the exponent of an irreducible f in lcm(g_a, h_a#) is max(v_f(g_a), v_f(h_a#)); on an orbit O, v_{f_i}(g_a) = α_i and v_{f_i}(h_a#) = β_i = P − α_{i−1}, giving the exponent max(α_i, P − α_{i−1}).

For the dimensions, Lemma RR-1(iii) applied to the monic divisors g_a and lcm(g_a, h_a#) of M gives dim_K C(a) = n − deg g_a and dim_K Hull_k(C(a)) = n − deg lcm(g_a, h_a#). Now n = deg M = Σ_{f∈F} P·deg f (S4), deg g_a = Σ_f a_f·deg f, and deg lcm(g_a, h_a#) = Σ_f max(a_f, β_f)·deg f. Grouping the factors by τ-orbits (constant degree on each orbit, Lemma RR-2(vi)):

```
dim_K C(a) = Σ_{f∈F} (P − a_f)·deg f = Σ_O deg(f_O) Σ_{i=0}^{L_O−1} (P − α_i),
dim_K Hull_k(C(a)) = Σ_{f∈F} [P − max(a_f, β_f)]·deg f.
```

For each orbit index i, with u = α_i and v = P − α_{i−1}, the elementary identity P − max(u, v) = min(P − u, P − v) (both sides equal P − u if u ≥ v and P − v otherwise) and P − v = α_{i−1} give P − max(α_i, P − α_{i−1}) = min(P − α_i, α_{i−1}). Substituting yields the displayed hull formula. The F_q statement follows from [K : F_q] = m_s applied componentwise (S8/A10 bookkeeping). ∎

Corollary (code and hull dimensions, factorwise form). With β_f denoting the exponent of f in h_a# (Theorem T3), dim_K C(a) = Σ_{f∈F} (P − a_f)·deg f and dim_K Hull_k(C(a)) = Σ_{f∈F} [P − max(a_f, β_f)]·deg f. This is the computation actually used in the cross-checks of §9.

---

7. Dependency map and hypothesis audit

```
Fact 0.1 (division ⇒ PID) ⟶ RR-1 (divisors, dimension) ⟶ T1
Fact 0.2 (unique factorization) ⟶ RR-1, RR-5
RR-2 (ρ¯ automorphism; # multiplicative; root transport; τ permutation)
        ├⟶ T2
        └⟶ T3 exponents (with RR-3, Prop. 4.2)
RR-3 (Euclidean dual ⟨h^♮⟩; NO separability) ⟶ Prop. 4.2 (D(C) = ρ(C^⊥), twist) ⟶ T3
RR-5 (lcm in K[x] and in the quotient) ⟶ T4 (with T1, T3)
```

Hypothesis audit (each entry: where it is first needed):

Hypothesis | First needed at | What breaks without it

λ ∈ K∗ | RR-3 (h_0 ≠ 0 from g_0 h_0 = −λ) | h^♮ and f# are undefined (division by a zero constant term). Example: f = x has no reciprocal of the form (S6). Such f never divides S here because S(0) = −μ ≠ 0. This is PC-3's missing hypothesis in the frozen T2 statement.

gcd(n_0, p) = 1 | (S4) (separability of S, hence multiplicity exactly P per distinct factor) | S may be inseparable; the parametrization by distinct factors with exponents in {0,…,P} is no longer the divisor lattice of M.

Compatibility (S3) | Prop. 4.2, last step; T3's "same modulus" conclusion; T4 | D(C(a)) = ⟨h_a#⟩ still holds (Theorem T3, first part), but in the modulus x^n − λ^{−p^{d−k}}; h_a# ∤ M in general, so Lemma RR-5(ii) does not apply in one quotient and the T4 lcm formula is not available. Necessity example: in GF(9) (d = 2), take k = 1 and λ of multiplicative order 8; then λ^{1 + 3} = λ^4 ≠ 1 (as λ^4 has order 2, i.e. λ^4 = −1), so λ is incompatible and indeed λ^{−p^{d−k}} = λ^{−3} ≠ λ. This shows the hypothesis cannot simply be dropped; it is a scope boundary, not a counterexample to the conditional theorems.

Dual order (S7) | Prop. 4.2 | With the opposite slot order, D(C) = σ(C^⊥) and the generator would be built from σ instead of ρ; the frozen h_a# formula would be false under that reading. This is the PC-2 ambiguity, resolved in this file by (S7) verbatim.

Orbit orientation (S8) | T3 exponents, T4 | Writing an orbit against the direction of τ replaces α_{i−1} by α_{i+1}; the formulas are not invariant under that relabelling.

Counterexample search outcome. No lemma or theorem above fails under (S1)–(S8). The three bullets in the table are necessity witnesses for hypotheses (what goes wrong outside the hypotheses), not counterexamples to the conditional statements. If a future command relaxes compatibility, T3 survives in the two-modulus form stated in Theorem T3, but T4 must be re-proved as a vector-space intersection across two different constacyclic presentations; the present file makes no claim there.

---

8. Remaining proof obligations (not part of this package)

1. **T5/T6** (transfer trace; global joint enumerator; multi-component CRT assembly; P = 1 recovery bookkeeping) — proof plan in P1_03_MATHEMATICAL_PROOF_PLAN.md §3, formal write-up pending.

2. **Corollaries C1–C4** as formal consequences of T6 (routes verified in P1-03; not written as proofs here); **C5** requires the unapproved PC-1 decision (frozen hull-mean is incorrect for L_O = 1); **C6** (RR-8 variance closed forms) is open; **C7** (RR-9) remains conditional on externally known distances.

3. **PC-1…PC-6** remain unapproved; in particular PC-2 (adding the (S7) sentence to the frozen convention) and PC-3 (adding f(0) ≠ 0 to frozen T2) are wording-level repairs awaiting a single approval command. This file adopts both **inside its own hypotheses only**.

4. **Gate G1** (literature/novelty) is untouched by this package and remains open (repo L1 theorem text unresolved; no novelty claim is made or implied by the correctness of these proofs).

---

9. Computational cross-check (supporting evidence, not proof)

Executed for this package (21 Oct 2026):

```
python3 code/validate_p1_03.py             # rerun for P1-04: exit 0, output identical to code/validation_p1_03_output.txt
                                  # final line: ALL P1-03 EXTENDED CHECKS PASSED
python3 /tmp/check_rr3.py         # RR-3-specific direct check (script kept in /tmp, not a repo artifact)
                                  # final line: ALL RR-3 DIRECT CHECKS PASSED
```

What was checked, and against which proof step:

- **RR-3 directly:** for GF(2) (27 assignments), GF(3) with λ = 1 (64) and λ = −1 (16), and GF(9), k = 1, with λ = 1 (256) and λ = −1 (256) — 619 assignments total — the shifts of h^↾ were verified to be Euclidean-orthogonal to every generator-basis row of C and to have the predicted dimension deg g, hence to equal C^⊥ in those cases, by the same inclusion-plus-dimension argument as the proof.
- **T3/T4 (via the P1-03 vector suite):** in all six vector cases (GF(2), GF(3)^{λ=±1}, GF(9)^{λ=±1}, GF(8) with k = 1, σ ≠ ρ, orbit lengths 1, 6; 2,806 assignments), the joint (dim_K C, dim_K Hull) distribution from the code-first Gram matrix dim Hull = dim C − rank(Σ_j B_{t,j} σ(B_{r,j})) agrees exactly with the T4 exponent formula of §6 and with the transfer product. The λ = −1 cases exercise the nontrivial-twist transport of Proposition 4.2; the GF(8) case exercises the ρ/σ distinction of (S7).

These checks are finite and single-component; they support the proved statements in the tested range (and, in P1-03, refuted the frozen C5 mean) but form no part of the proofs above.